Rakathon 2026 · FN Motol · Digital adverse-event watchdog

Digital Guardian

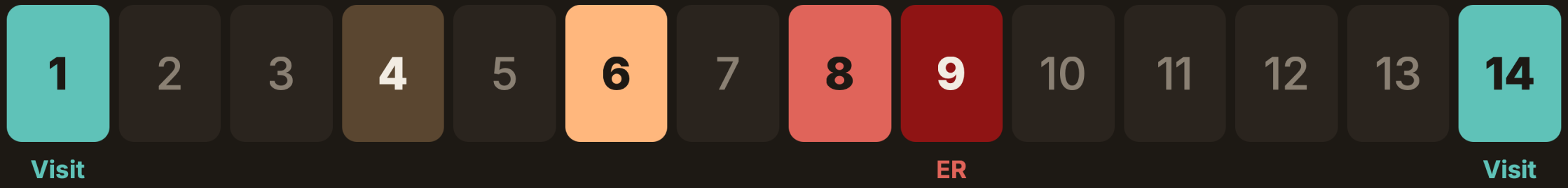
Every cancer patient, heard between visits.







Meet Jan. 66. Kidney cancer. New pill on day 1.



Day 4

A little diarrhea. “Probably nothing.”

Day 6

Worse. He doesn’t want to bother anyone.

Day 8

He can’t leave the bathroom.

Day 9

Emergency room, dehydrated. His next visit was day 14.

Jan is a fictional patient. Every patient in our demo is synthetic.

The problem

In between, the clinic hears nothing.

For patients

14–21 **days of silence**
Side effects reach the clinic only if the patient calls.

14–17 **questions per full check-in**
Too many for every day. Small problems go unreported.

41% **ended up in the ER**
on usual care during chemotherapy (trial, 766 patients).

For doctors

Unprepared for the visit
Two to three weeks of symptoms, pieced together from memory in a short appointment.

Buried in notifications
If every minor symptom raises an alarm, precious time goes to noise instead of the patients who need it.

Cycle lengths: official drug labels. ER visits: Basch et al., J Clin Oncol 2016 (single centre; abstract and secondary sources). Cited evidence, not tested by our prototype.

Our solution

Digital Guardian listens between visits and speaks up only when it matters.

5/17

questions per check-in. It stops as soon as the answer can't change.

Red

notifies at once. Yellow and grey wait quietly in the worklist.

0

cases rated too mild or too harsh in 20,000 simulated check-ins.

1 click

from an alert to its rule, values and evidence. The doctor is prepared before the visit.

Why listening matters

In the same trial, patients who reported symptoms regularly lived longer: median **31.2 vs 26.0 months.**

Our numbers: simulated sessions with synthetic answers and placeholder rules, 8 therapy profiles. "1 click": tested end to end in the prototype. Trial: Basch et al., JAMA 2017 (abstract and secondary sources); cited evidence, not tested by our prototype.

Live demo · synthetic data

View the demo

Safety by design · EU AI Act principles

Patient data stays protected.

Data protection

- Synthetic data only, no real patients in code or logs
- AI in the EU data zone, with storage switched off
- Pseudonyms in every export, only the birth year
- From the clinic’s dataset: aggregate counts only
- Patients see only their own data (server-enforced)

Human oversight

A doctor acknowledges every alert and sets every grade.

Transparency

Every alert shows its rule and values; AI text is sourced.

Record-keeping

Every acknowledgement, export and access is logged.

Accuracy

Every rule version is checked on every case.

Research prototype on synthetic data. Designed along the principles of the EU AI Act, MDR and GDPR; no conformity assessment, not certified.
Model: gpt-6.1-sol via the Rakathon gateway, store=false.

Why not just a chatbot?

A chatbot can sound right. Medicine needs to be right.

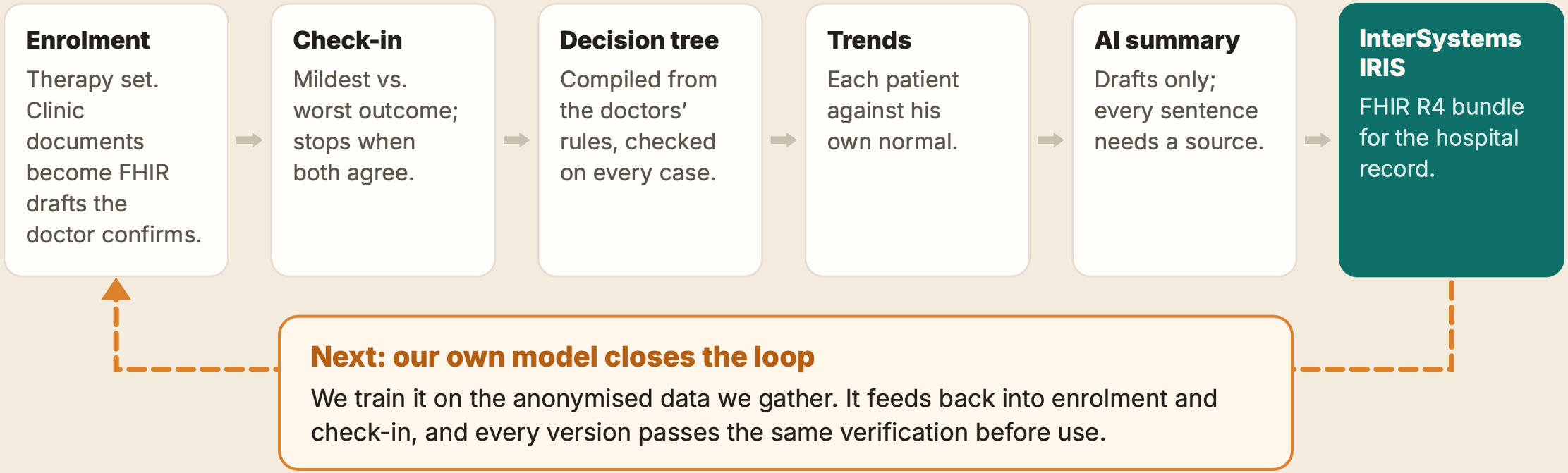
	A plain chatbot	Digital Guardian
Who sets the alert level?	The model, possibly differently each time	Verified rules: same answers, same level
Can it make things up?	Yes, it can hallucinate	A checker deletes every sentence without a source
Where does the data go?	Possibly stored, possibly used for training	EU data zone, storage off, synthetic data only
And if the model fails?	Nothing works	A fixed template takes over; alerts still work

We still use AI where it is strong: it **extracts facts** from clinic documents into HL7 FHIR drafts and **writes** the summary. Every result is checked against its source.

Model: gpt-6.1-sol via the Rakathon gateway, EU data zone, store=false. In our runs the checker dropped up to a third of the Czech summary sentences.

The engineering

One full cycle, and it learns.



Today the tree is compiled only from the doctors' rules, with no patient data; our own model is the next step, not built yet. Document import is a proof of concept. IRIS push tested against a local FHIR server only.

Engineering · deciding safely

Doctors write the rules. We check every single case.

~2 million

answer combinations per therapy profile, every one checked: the tree equals the rulebook

0

cases rated too mild, in any combination and in 20,000 simulated check-ins

0

cases rated too harsh, under the same checks

Locked safety floors: fever is always red. Every result explains itself: “would drop if diarrhea fell to grade 2 or less.” And the tree grows: new therapies plug in as modules that can only tighten, never loosen.

Faithful implementation of placeholder rules, not clinical validation. 8 therapy profiles. The tree is compiled from rules, not trained on patient data.

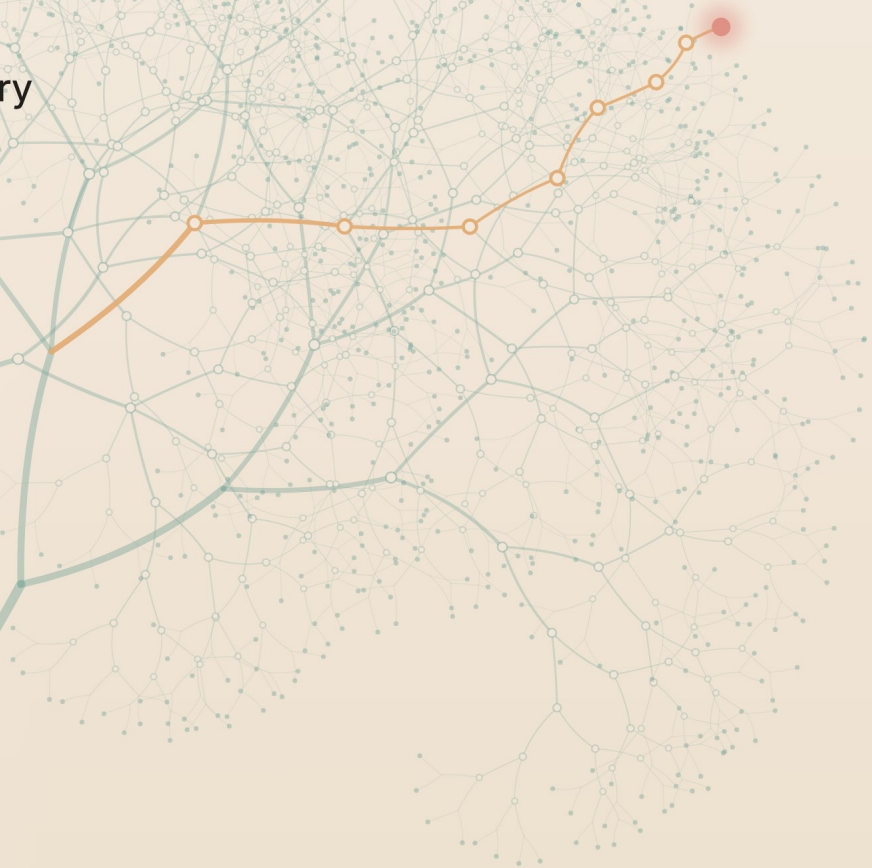
What comes next

Built to grow.

New data. Clinic documents, drug labels, the patient's own words. Every fact traced.

New therapies. Modules plug into the tree without new code, verified before use.

Our own model. Learns from anonymised outcomes. Never less safe than the rules.



Today it hears Jan.
Tomorrow it learns from every Jan.